Sales Analysis

using sql

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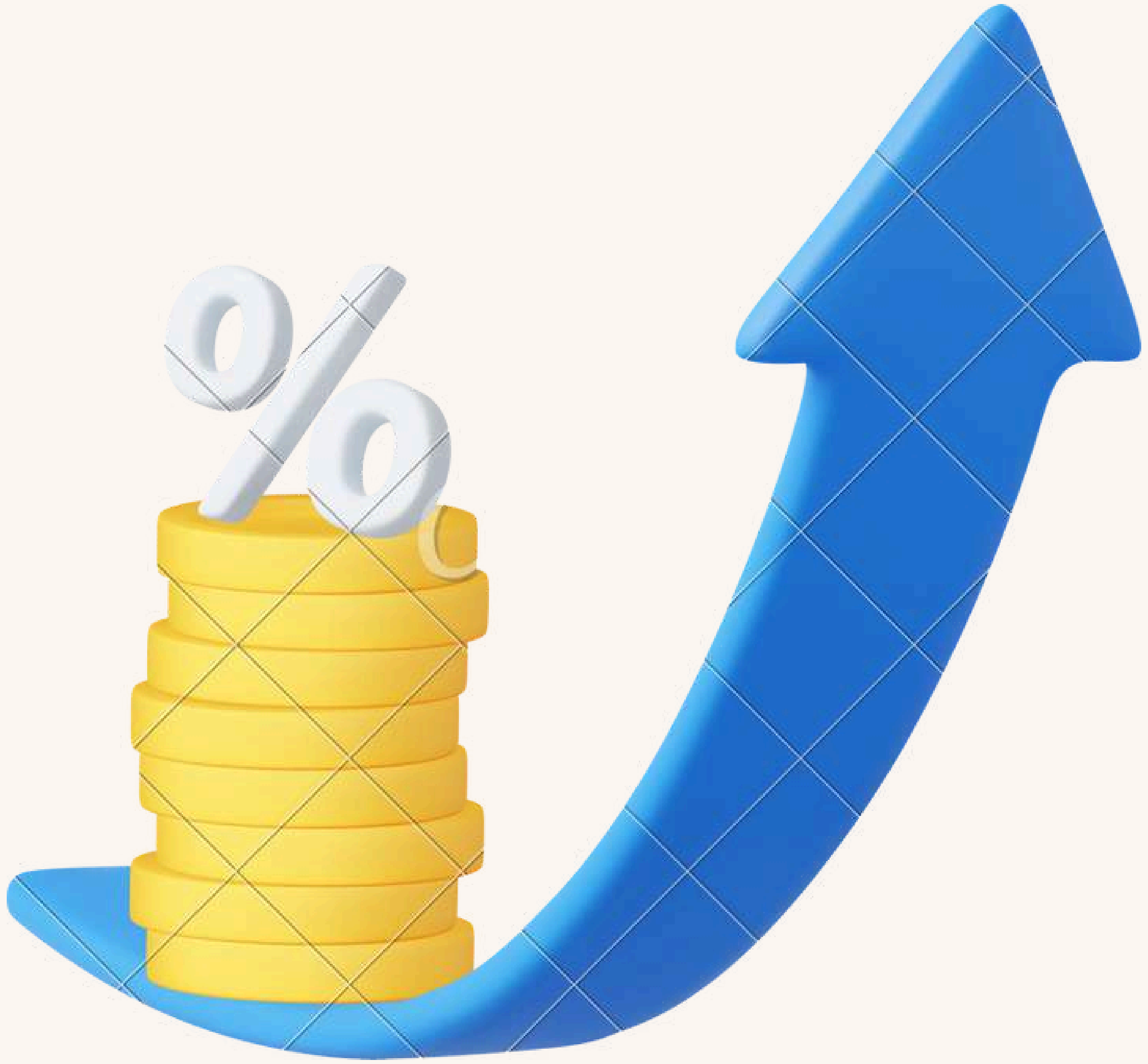
Project Overview

The project focuses on sales data analysis and customer relationship management(CRM) analysis over a one-year period to derive actionable insights using SQL. The project goal is to derive insights of
sales patterns, customer behaviors and category trends.

Objectives of this project

- Analyze sales performance
- Understand customer spending habits.
- Evaluate regional sales trends
- assess category performance.



Methodology

1. Data Collection and Preparation

Datasets Used:

- Order_Details: Contains detailed order information (order ID, quantity, amount, , category. subcategory)
- Order_Summary: Summarizes orders (order ID, order date, customer name, state, city).
- Sales_Target: Includes sales targets for each month and each category(date, category, target)

Data Cleaning:

- Tools Used: Excel

2. Sql analysis

- Analyze sales performance-Calculated key metrics such as total revenue and average order value
- Evaluate regional and customer sales trends-Analyzed sales data to identify trends by region and customer spending habits
- Assess category performance-Evaluated sales performance across different product categories .

3. Documenting

Compiled and summarized key findings.

Created comprehensive documentation detailing the analysis process, findings, and actionable insights.

Sales analysis

Order Per Day:

select os.order_date, sum(od.amount*od.quantity) **from**
order_summary os **inner join** order_details od
on os.order_id=od.order_id
group by os.order_date
order by os.order_date;

Total Sum of Quantities Sold Under Each Order ID:

select a.order_id ,sum(a.quantity) **from**
order_details a **inner join** order_summary b on
a.order_id=b.order_id
group by a.order_id
order by order_id;

highest and lowest quanity sold

```
(select sum(quantity), category from order_details
 group by category
 order by sum(quantity) desc
 limit 1)
 union all(
 select sum(quantity), category from order_details
 group by category
 order by sum(quantity)
 limit 1);
```

sum	category
3516	Clothing
945	Furniture

Difference Between Monthly Target and Monthly Sales by Each Category

```
select st.order_date,sq.months,st.category,sq.monthly_total_sales,st.target,st.target-sq.monthly_total_sales as difference
 from sales_target st left join (
select a.category, sum(a.amount*a.quantity) as monthly_total_sales ,extract (month from b.order_date) as months
 from order_details a inner join order_summary b
on a.order_id=b.order_id
 where extract(month from b.order_date) in (select extract(month from st.order_date) from sales_target st)
group by extract (year from b.order_date),months,a.category )sq  on extract(month from st.order_date)=sq.months
and st.category =sq.category
order by extract (year from st.order_date), sq.months,st.category;
```

Identify Orders Where the Target Wasn't Met

```
select sbq.months,st.target,sbq.monthly_sales,sbq.category from sales_target st left join (
select sum(a.amount*a.quantity) as monthly_sales, extract(month from b.order_date) as months,a.category
from order_summary b inner join order_details a
on a.order_id=b.order_id
where extract(month from order_date) in (select extract(month from st.order_date) from sales_target st)
group by months,a.category) as sbq on extract(month from st.order_date) = sbq.months and st.category=sbq.category
where st.target>sbq.monthly_sales
order by sbq.months;
```

months	target	monthly_sales	category
7	14000	13060.00	Clothing

calculate the percentage of the highest monthly sales compared to the total yearly sales

```
select round((bc.total_sales/ab.total_sales_yearly)*100) from (select sum(amount*quantity) as
total_sales_yearly
 from order_details) ab cross join
(select sum(a.amount*a.quantity) as total_sales, extract(month from b.order_date) as months
 from order_details a inner join order_summary b on a.order_id=b.order_id
   group by months
   order by total_sales desc
 limit 1) bc ;
```

"round"
16

CRM Analysis

List the top 3 customers with the highest total purchase amount for each month

```
select customer_name,total_sales,years,months, sales_rank from(
select os.customer_name , sum(od.amount*od.quantity) as total_sales,extract(year from os.order_date)
as years ,extract(month from os.order_date)
as months,rank()
over(partition by extract(month from os.order_date) order by sum(od.amount*od.quantity) desc ) as
sales_rank
from order_summary os inner join order_details od
on os.order_id=od.order_id
group by years,months,os.customer_name,os.state,os.city)
where sales_rank <=3
order by years,months,sales_rank;
```

how many customers purchased more than once

```sql
select a.customer_name, count( distinct (b.order_id)) as frq
from order_details b inner join order_summary a on a.order_id=b.order_id
group by a.customer_name,a.state,a.city
having count( distinct (b.order_id))>1
order by frq desc;
```

top 3 customers based on their purchases

```sql
select os.customer_name, sum(od.amount*od.quantity) as total_order
from order_summary os inner join order_details od on os.order_id=od.order_id
group by os.customer_name,os.state,os.city
order by total_order desc
limit 3
```

customer_name	total_order
Yaanvi	103435.00
Seema	64222.00
Soumya	42906.00

Top 3 Highest Sales Cities

```
select city from (
select b.city,sum(a.amount*a.quantity) as monthly_sales from order_details a inner join order_summary b
on a.order_id=b.order_id
group by b.state,b.city
order by monthly_sales desc)
limit 3;
```

"city"
"Indore"
"Mumbai"
"Pune"

lowest sale city

```
select b.city,sum(a.amount*a.quantity) as monthly_sales
from order_details a inner join order_summary b
on a.order_id=b.order_id
group by b.state,b.city
order by monthly_sales
limit 1;
```

city	monthly_sales
Delhi	8966.00

Category Analysis

monthly sales by each category

```
select a.category, sum(a.amount*a.quantity) as ts,
extract (year from b.order_date) as years ,
extract(month from b.order_date) as months from order_details a inner join order_summary b
on a.order_id=b.order_id
group by rollup (years, months,a.category)
order by years, months,a.category;
```

which category has generated the highest revenue

```
select sum(amount*quantity) as total_sales,
category from order_details
group by category
order by total_sales desc
limit 1;
```

total_sales	category
816583.00	Electronics

which category has highest sales in each month

```
select category, total_sales,years,months from (
select a.category, sum(a.amount*a.quantity) total_sales,
 extract (year from b.order_date) as years ,
 extract(month from b.order_date) as months,
   rank() over(partition by extract(month from b.order_date)
    order by sum(a.amount*a.quantity) desc)as ranks
   from order_details a inner join order_summary b
   on a.order_id=b.order_id
   group by(years, months,a.category))
   where ranks=1
   order by years, months ;
```

category	total_sales	years	months
Clothing	67649.00	2018	4
Electronics	61765.00	2018	5
Electronics	43234.00	2018	6
Electronics	22939.00	2018	7
Clothing	59474.00	2018	8
Clothing	58752.00	2018	9
Electronics	69151.00	2018	10
Electronics	92824.00	2018	11
Electronics	92979.00	2018	12
Electronics	161965.00	2019	1
Furniture	79781.00	2019	2
Furniture	130043.00	2019	3

Identify the month with the highest sales for each category in the year

```
select category,months  from(
select extract(month from b.order_date)as months,
a.category,sum(a.amount*a.quantity) as total_sales_pm,
rank() over(partition by category order by sum(a.amount*a.quantity)desc )  as ranks
from order_details a inner join order_summary b on a.order_id=b.order_id
group by a.category,months)
where ranks=1;
```

category	months
Clothing	3
Electronics	1
Furniture	3

Determine the category with the highest growth in sales compared to the previous month

```
select category,max(next_ms-ts)as growth from(
select category, ts,lead(ts) over( partition by category order by category ) as next_ms,months from(
select a.category,
 sum(a.amount*a.quantity) as ts,extract (year from b.order_date) as years,
 extract(month from b.order_date) as months from order_details a inner join order_summary b
   on a.order_id=b.order_id
   group by years, months,a.category
   order by a.category,years, months) )
   group by category
   order by growth desc
   limit 1;
```

category	growth
Furniture	79775.00

KEY FINDINGS

SALES OVERVIEW:

- Total sales for the period Apr 2018 to Mar 2019 - Rs 2146870.00
- Highest revenue month -Jan 2019 of Rs 337229.00 (16% of total sales)
- The month and category in which sales target was not met - clothing in 7th month of 2018.
- Highest growth in sales compared to last month - furniture category in Jan 2019 (growth -2.8 times)

CATEGORY PERFORMANCE:

- Highest revenue category- Electronics
  amount generated-Rs 816583 (38% of total sales.)
- Highest quantity sold - clothing
- Lowest quantity sold - furniture
- Highest revenue generated (month and city)
  Clothing - march 2019, Indore
  Electronics - Jan 2019 ,Indore
  Furniture -March 2019.,Indore

CUSTOMERS AND GEOGRAPHIC INSIGHTS:

- Top 3 cities with highest sales
  Indore
  Mumbai
  Pune
- City with lowest sale
  Delhi
- Top 3 customers according to their purchases
  Yaanvi
  Seema
  Saumya
- No. of customers who purchased more than once-70

CONCLUSION

In conclusion, the sales data from April 2018 to March 2019 highlights a decent performance of the business as sales targets were surpassed in almost every month with Jan month witnessing the highest sales. Key highlights include significant revenue contributions from the Electronics category. Cities like Indore, Mumbai, Pune has the highest sales and attention should be given to cities with lower sales performance like Delhi to identify opportunities for growth.

Business also has a loyal customer base which is beneficial for the business and to sustain this performace or to enhance it addressing underperforming areas, and improving customer retention strategies would be beneficial.

Thank you very much!